

NBACore v8.3.10

Competition Rules Engine

AI Development Work Contract

Version

v8.3.10

Module Type

Basketball Context Infrastructure

Execution Target

WorkBuddy / Codex / Local AI Coding Agent

1. Module Objective

建立 NBACore 比赛规则环境系统。

目标：

让系统理解：

不同篮球比赛环境下：

- 数据统计方式
- 比赛规则
- 球场条件
- 时间规则
- 技术规则

对球员表现产生的影响。

2. Core Philosophy

篮球数据不是脱离环境存在。

错误模型：

```text id="f6c8hm" Player

↓

Statistics

正确模型:

```text id="5g0p4m"

Player

↓

Performance

↓

Competition Environment

↓

Ruleset

3. Problem Definition

不同篮球体系存在差异。

例如:

NBA

```text id="q9sg1s" Quarter:

12 minutes

Game:

48 minutes

Three Point:

7.24m

Defensive 3 seconds:

YES

---

## FIBA

```text id="36r7tm"

Quarter:

10 minutes

Game:

40 minutes

Three Point:

6.75m

Defensive 3 seconds:

NO

NCAA

```text id="8y7j4t" Half:

20 minutes

Three Point:

different by era

Shot Clock:

different by era

---

# 4. Design Principle

Rules Engine 不修改原始数据。

保存：

```
```text id="wrv8ri"
```

Raw Statistics

+

Context Metadata

例如：

球员：

```
```text id="4h2yfk" 30 PPG
```

系统保存：

```
```json id="4t1q7m"
```

```
{
```

```
"value":30,
```

```
"competition":"NBA",
```

```
"ruleset":"NBA_2026"
```

```
}
```

5. Architecture Position

完整数据流：

```
```text id="4wxyy6" Import
```

↓

Ontology Layer

↓

Provenance Layer

↓

Competition Rules Engine

↓

Data Asset System

↓

Formula Engine

↓

Visualization

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## # 6. Core Objects

系统包含：

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### # 6.1 Competition

代表比赛环境。

例如：

NBA Regular Season

```
```json id="jzv6dx"
{
  "id": "nba",
  "name": "NBA",
  "country": "USA",
  "level": "professional"
}
```

6.2 Ruleset

代表具体规则版本。

例如：

NBA 2025-26

```
```json id="r4f3ja" {
```

```
"id": "nba_2025",
```

```
"quarter_length": 12,
```

```
"game_length": 48,
```

```
"shot_clock": 24
```

```
}
```

---

# 6.3 Era

规则时代。

因为规则会变化。

例如：

NBA：

1990s

2000s

2020s

---

# 6.4 Competition Context

比赛记录关联环境。

例如:

```
```json id="0f2y4k"
{
  "game_id":123,
  "competition":"NBA",
  "ruleset":"nba_2026"
}
```

7. Database Design

7.1 Competition Table

```
```sql id="0m2w3x" CREATE TABLE competitions(
 id INTEGER PRIMARY KEY,
 name TEXT,
 country TEXT,
 level TEXT
);
```

---

## 7.2 Ruleset Table

```
```sql id="d3hn5d"
CREATE TABLE rulesets(
  id INTEGER PRIMARY KEY,
  competition_id INTEGER,
  version TEXT,
  quarter_length INTEGER,
  shot_clock INTEGER,
```

```
three_point_distance FLOAT  
  
);
```

7.3 Era Table

```
```sql id="9qdq9m" CREATE TABLE competition_eras(  

id INTEGER PRIMARY KEY,

competition_id INTEGER,

start_year INTEGER,

end_year INTEGER,

ruleset_id INTEGER

);
```

```

```

### # 8. Backend Structure

创建:

```
```text id="k5xh8s"  
backend/  
  
services/  
  
competition/  
  
    manager.py  
  
    rules.py  
  
    context.py
```


9. Core Classes

CompetitionManager

```python id="pkp2cn" class CompetitionManager:

```
def create_competition():
 pass

def get_ruleset():
 pass

def compare_environment():
 pass
```

```

10. Rules Definition

规则字段：

Time Rules

```text id="7d2d3m"  
quarter_length  
  
game_length  
  
overtime_length
```

Court Rules

```text id="0r1d5c" three\_point\_distance

court\_size

paint\_area

```

Tactical Rules

```text id="9s82nz"  
defensive_three_seconds  
  
zone_defense  
  
goaltending
```

Statistical Rules

```text id="j6g8p1" possessions

pace\_calculation

stat\_definition

```

11. API Requirements

Get Ruleset

GET:
```

/api/competition/{id}/ruleset

```
返回:

```json id="4sx6tv"  
{  
  
  "quarter_length":12,  
  
  "shot_clock":24  
  
}
```

Compare Competition

GET:

```
/api/competition/compare
```

输入:

```
```json id="br7x8k" {  

 "competition_a":"NBA",

 "competition_b":"FIBA"

}
```

---

返回:

```
```json id="p8j7vc"  
{  
  
  "differences":[  
  
    "quarter_length",  
  
    "three_point_distance"  
  
  ]  
  
}
```

12. Formula Engine Integration

公式计算时:

可以读取环境。

例如:

普通:

PTS

高级：

PTS adjusted by competition

未来支持：

```text id="y2s4hc" NBA PTS

↓

FIBA Context Adjusted PTS

---

# 13. Player Comparison Integration

支持：

球员跨环境分析。

例如：

球员：

```text id="f8l4cz"

NBA：

25 PPG

FIBA：

21 PPG

系统显示：

环境差异：

- 比赛时间
- 节奏
- 三分距离

14. Frontend Requirements

第一阶段：

提供：

Competition Selector

例如：

League:

[NBA ▼]

Season:

[2025-26 ▼]

Rules:

[Current Rules]

比较页面：

NBA

VS

FIBA

显示：

规则差异。

15. Testing Requirements

Unit Test

测试：

创建规则

读取规则

版本切换

Integration Test

测试：

NBA数据导入

↓

关联NBA Ruleset

FIBA数据导入

↓

关联FIBA Ruleset

16. Acceptance Criteria

完成标准：

系统可以：

✔ 保存不同篮球规则

✔ 区分比赛环境

✔ 支持规则版本变化

✔ 支持跨联赛比较

✔ 不破坏原始统计

17. Development Notes For AI Agent

重要原则：

不要：

```
if league=="NBA":
```

硬编码。

必须：

```
ruleset = CompetitionManager.get_ruleset()
```

不要：

认为：

NBA数据 = 篮球全部数据。

必须：

保持：

```text id="3r3q8c" Player

+

Competition Context

```

18. Future Extension

未来支持：

Rule Impact Model

分析：

规则变化如何影响球员。
```

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## ## Cross Competition Translation

例如：

NBA表现：

↓

FIBA环境预测

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## ## AI Context Analysis

AI解释：

为什么同一个球员在不同联赛表现不同。

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## # Final Principle

NBACore 不比较孤立数字。

NBACore 比较：

```text id="qk0hjp"

Player

+

Competition

+

Rules

+

Context

这才是真正的篮球智能分析。

END